Sample Program Guide

Revision. 1.2

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Introduction

This guide explains how to use this sample program.

Usage

Environment

The sample program is a command-line program and runs on the following environments.

- Operating environment
- Contents - [Windows] - Sample Program
- Contents - [Macintosh] - Sample Program

Files

The sample program needs following files. Copy these files into the appropriate folder and use them.

Windows

File Name	Folder
ControlServiceLayer.dll	Same folder as this sample program
NkdPTP.dll	Same folder as ControlServiceLayer.dll
dnssd.dll	Same folder as ControlServiceLayer.dll
NkRoyalmile.dll	Same folder as ControlServiceLayer.dll
DC_PTP_Config.config	C:\Users\<User>\AppData\Local\Nikon\NXTether
MaidLayer.config	C:\Users\<User>\AppData\Local\Nikon\NXTether
RangeValue.config	C:\Users\<User>\AppData\Local\Nikon\NXTether

Macintosh

File Name	Folder
TypeCommon Module.bundle	
libNkPTPDriver2.dylib	
libMCARecLib3.dylib	
Royalmile.framework	
DC_PTP_Config.config	~/Library/Preferences/Nikon/NXTether
MaidLayer.config	~/Library/Preferences/Nikon/NXTether
RangeValue.config	~/Library/Preferences/Nikon/NXTether

Main Menu
When you start the sample program, it loads ControlService. Then main menu is shown.

Select (1, 0)
1. Select Device
If some devices are connected to PC, you can choose which device will be used.
0. Exit
You can exit from current menu.

After selecting a device, the menu below will be shown
Select (1-11, 0)

1. Camera Settings (1)	2. Camera Settings (2)	3. Shooting Menu
4. Live View	5. Autofocus	6. Capture
7. CaptureAsync	8. AFCaptureAsync	9. AFCapturesync
10. SB Menu	11. MovieMenu	0. Exit

1. Camera Settings (1)

You can change some camera settings. Sub menu will be display as in below list.
1)
Select the item you want to set up.

1. IsAlive	2. Name	3. Interface
4. DataTypes	5. BatteryLevel	6. LockExposure
7. ExposureMode	8. ShutterSpeed	9. Aperture
10. ExposureComp	11. SaveMedia	12. MovieWBAutoType
14. MovieAperture	15. MovieExposureComp	
19. LensType	21. MovieShutterSpeed	24. EnableBracketing
25. BracketingOrder	26. AEBracketingStep	27. WBBracketingStep
28. BracketingType	29. ADLBracketingType	30. ADLBracketingStep
31. BracketingCount	32. BracketingVary	0. Exit

1. IsAlive
Show the status of Module object.
(refer to kNkMAIDCapability_IsAlive in MAID3-COMMON(E).PDF)

2. Name
Show the name of Module object.
(refer to kNkMAIDCapability_Name in MAID3-COMMON(E).PDF)

3. Interface
Show the physical interface of this Source object.
(refer to kNkMAIDCapability_Interface in MAID3-COMMON(E).PDF)

4. DataTypes
Show the data types supported by this Item object.
(refer to kNkMAIDCapability_DataTypes in MAID3-COMMON(E).PDF)

5. BatteryLevel
Show battery level.
(refer to kNkMAIDCapability_BatteryLevel in MAID3-COMMON(E).PDF)

6. LockExposure
Show the status of exposure lock.
(refer to kNkMAIDCapability_LockExposure in MAID3-COMMON(E).PDF)

7. ExposureMode
Set the exposure mode of camera.
(refer to kNkMAIDCapability_ExposureMode in MAID3-COMMON(E).PDF)

8. ShutterSpeed
Set the exposure time of camera.
(refer to kNkMAIDCapability_ShutterSpeed in MAID3-COMMON(E).PDF)

9. Aperture
Set the aperture of camera.
(refer to kNkMAIDCapability_Aperture in MAID3-COMMON(E).PDF)

10. ExposureComp
Set the exposure compensation of camera.
(refer to kNkMAIDCapability_ExposureComp in MAID3-COMMON(E).PDF)

11. SaveMedia
This will set the recording media.
(refer to kNkMAIDCapability_SaveMedia in MAID3-COMMON(E).PDF)

12. MovieWBAutoType
This will set the option when the white balance auto. (Video Recording Menu).
(refer to kNkMAIDCapability_MovieWBAutoType in MAID3-COMMON(E).PDF)

14. MovieAperture
This will set the aperture during movie live view.
(refer to kNkMAIDCapability_MovieAperture in MAID3-COMMON(E).PDF)

15. MovieExposureComp
This will set the exposure compensation for movie live view.
(refer to kNkMAIDCapability_MovieExposureComp in MAID3-COMMON(E).PDF)

19. LensType
Shows the lens type about CPU lens.
(refer to kNkMAIDCapability_LensType in MAID3-COMMON(E).PDF)

21. MovieShutterSpeed
This will set the shutter speed during movie live view.
(refer to kNkMAIDCapability_MovieShutterSpeed in MAID3-COMMON(E).PDF)

24. EnableBracketing
This will set whether bracketing is applied or not.
(refer to kNkMAIDCapability_EnableBracketing in MAID3-COMMON(E).PDF)

25. BracketingOrder
This will set the bracketing order. (Custom setting menu e7).
(refer to kNkMAIDCapability_BracketingOrder in MAID3-COMMON(E).PDF)

26. AEBracketingStep
This will set the exposure increment for AE, SB, AE/SB bracketing.
(refer to kNkMAIDCapability_AEBracketingStep in MAID3-COMMON(E).PDF)

27. WBBracketingStep
This will set the white balance increment for WB bracketing.
(refer to kNkMAIDCapability_WBBracketingStep in MAID3-COMMON(E).PDF)

28. BracketingType
This will set the combination bracketing shots and direction when AE, Flash, AE and Flash, White balance bracketing.
(refer to kNkMAIDCapability_BracketingType in MAID3-COMMON(E).PDF)

29. ADLBracketingType
This will set the bracketing shots when ADL bracketing.
(refer to kNkMAIDCapability_ADLBracketingType in MAID3-COMMON(E).PDF)

30. ADLBracketingStep
This will set ADL bracketing level when two shots ADL bracketing is selected.
(refer to kNkMAIDCapability_ADLBracketingStep in MAID3-COMMON(E).PDF)

31. BracketingCount
This will get the number of shots on AE bracketing or ADL bracketing.
(refer to kNkMAIDCapability_BracketingCount in MAID3-COMMON(E).PDF)

32. BracketingVary
Select a Bracketing variation.
(refer to kNkMAIDCapability_BracketingVary in MAID3-COMMON(E).PDF)

2. Camera Settings (2)

You can change other camera settings. Sub menu will be display as in below list.

2)

Select the item you want to set up

1. LockCamera	2. LensInfo	3. IsoControl
4. ISOControlSensitivity	5. MovieISOControl	6. MovieSensitivity
9. Active_D_Lighting	10. CCDDataMode	11. CompressionLevel
13. CompressRAWEx	14. PortraitImpressionBalance	15. FlashISOAutoHighLimit
16. SkinSoftening	17. FocalLength	18. FocusAreaMode

19. FocusMode
20. ToneMode
21. VibrationReduction
22. Viewmode
23. MeteringMode
24. ISOAutoHiLimit
25. ISOAutoShutterTime
26. PictureControl
27. PictureControlHlg
28. ChangeMonitorOffStatus
0. Exit

1. LockCamera
Keep the camera from direct operation (not through host computer).
(refer to kNkMAIDCapability_LockCamera in MAID3-COMMON(E).PDF)

2. LensInfo
Show lens information (focal length and aperture).
(refer to kNkMAIDCapability_LensInfo in MAID3-COMMON(E).PDF)

3. IsoControl
Set whether auto sensitivity control is used when you take a picture.
(refer to kNkMAIDCapability_IsoControl in MAID3-COMMON(E).PDF)

4. ISOControlSensitivity
This will get the ISO sensitivity which the camera is controlling.
(refer to kNkMAIDCapability_ISOControlSensitivity in MAID3-COMMON(E).PDF)

5. MovieISOControl
This will set “Movie ISO sensitivity setting - Auto ISO control (mode M)”. (Video Recording Menu).
(refer to kNkMAIDCapability_MovieISOControl in MAID3-COMMON(E).PDF)

6. MovieSensitivity
This will set “ISO sensitivity (mode M)” for movie live view. (Video Recording Menu).
(refer to kNkMAIDCapability_MovieSensitivity in MAID3-COMMON-Z6_3(E).PDF)

9. Active_D_Lighting
This will set Active D-Lighting. (Photo shooting menu).
(refer to kNkMAIDCapability_Active_D_Lighting in MAID3-COMMON(E).PDF)

10. CCDDataMode
This will set the value of [Image area] - [Choose image area]. (Photo shooting menu).
(refer to kNkMAIDCapability_CCDDataMode in MAID3-COMMON(E).PDF)

11. CompressionLevel
Set compression level of the image file created by camera.
(refer to kNkMAIDCapability_CompressionLevel in MAID3-COMMON(E).PDF)

13. CompressRAWEx
This will set the value of [NEF(RAW) recording] – [NEF(RAW) compression]. (Photo shooting menu).
(refer to kNkMAIDCapability_CompressRAWEx in MAID3-COMMON(E).PDF)

14. PortraitImpressionBalance
This will set Portrait Adjustment" in the still image shooting menu.
(refer to kNkMAIDCapability_PortraitImpressionBalance in MAID3-COMMON(E).PDF)

15. FlashISOAutoHighLimit
This will set the "ISO sensitivity setting - Automatic sensitivity control - Maximum control sensitivity when using flash" in the still image shooting menu.
(refer to kNkMAIDCapability_FlashISOAutoHighLimit in MAID3-COMMON(E).PDF)

16. SkinSoftening
This represents the "beautiful skin effect" option in the still image shooting menu.
(refer to kNkMAIDCapability_SkinSoftening in MAID3-COMMON(E).PDF)

17. FocalLength
Show focal length (unit: mm).
(refer to kNkMAIDCapability_FocalLength in MAID3-COMMON(E).PDF)

18. FocusAreaMode
Set focus area mode.
(refer to kNkMAIDCapability_FocusAreaMode in MAID3-COMMON(E).PDF)

19. FocusMode
Show focus mode.
(refer to kNkMAIDCapability_FocusMode in MAID3-COMMON(E).PDF)

20. ToneMode
Show tone mode.
(refer to kNkMAIDCapability_ToneMode in MAID3-COMMON(E).PDF)

21. VibrationReduction
This will set “Vibration reduction” of Photo Shooting Menu.
(refer to kNkMAIDCapability_VibrationReduction in MAID3-COMMON(E).PDF)

22. Viewmode
Set “View mode (photo Lv)”.
(refer to kNkMAIDCapability_ViewMode in MAID3-COMMON(E).PDF)

23. MeteringMode
Set the metering mode of camera.
(refer to kNkMAIDCapability_MeteringMode in MAID3-COMMON(E).PDF)

24. ISOAutoHiLimit
This will set "ISO sensitivity setting - automatic sensitivity control - maximum control sensitivity" in the still image shooting menu.
(refer to kNkMAIDCapability_ISOAutoHiLimit in MAID3-COMMON(E).PDF)

25. ISOAutoShutterTime
This will set the shutter speed when ISO is controlled automatically. (Photo shooting menu).

(refer to kNkMAIDCapability_ISOAutoShutterTime in MAID3-COMMON(E).PDF)

26. PictureControl
Set and get current Picture Control.
(refer to kNkMAIDCapability_PictureControl in MAID3-COMMON(E).PDF)

27. PictureControlHlg
Set the Picture Control (HLG)" in the still image shooting Menu.
(refer to kNkMAIDCapability_PictureControlHlg in MAID3-COMMON(E).PDF)

28. ChangeMonitorOffStatus
Switch the forced lighting off status on the camera's monitor and also show the camera's monitor status.
(refer to kNkMAIDCapability_ChangeMonitorOffStatus in MAID3-COMMON(E).PDF)

3. Shooting Menu

You can change photography settings. Sub menu will be display as in below list.
3)
Select the item you want to set up

1. WBMode	2. Sensitivity	3. WBFluorescentType
4. WBAutoType	5. WBTuneColorTempEx	6. MovieWBMode
7. MovieWBFluorescentType	8. ShootingMode	9. ContinuousShootingNum
10. RemainContShooting	11. ShootingLimit	12. RemainCountInMedia
13. MovieWBTuneColorTempEx	14. WBTuneColorAdjust	15. MovieWBTuneColorAdjust
21. SpecialShootingMode	22. Slot2JpegImageSize	0. Exit

1. WBMode
Set white balance mode(auto/Incandescent/...).
(refer to kNkMAIDCapability_WBMode in MAID3-COMMON(E) .PDF)

2. Sensitivity
Set sensitivity of camera.
(refer to kNkMAIDCapability_Sensitivity in MAID3-COMMON(E).PDF)

3. WBFluorescentType
Set the fluorescent type when the white balance is "Fluorescent".
(refer to kNkMAIDCapability_WBFluorescentType in MAID3-COMMON(E).PDF)

4. WBAutoType
Set white balance Auto type in Auto mode.
(refer to kNkMAIDCapability_WBAutoType in MAID3-COMMON(E).PDF)

5. WBTuneColorTempEx
This will set the color temperature when the WBMode is "Color Temperature". (Photo shooting menu).
(refer to kNkMAIDCapability_WBTuneColorTempEx in MAID3-COMMON(E).PDF)

6. MovieWBMode
This will set the white balance mode. (Video Recording Menu).
(refer to kNkMAIDCapability_MovieWBMode in MAID3-COMMON(E).PDF)

7. MovieWBFluorescentType
This will set the fluorescent type when the WBMode is “Fluorescent”.
(refer to kNkMAIDCapability_MovieWBFluorescentType in MAID3-COMMON(E).PDF)

8. ShootingMode
Set the shooting mode.
(refer to kNkMAIDCapability_ShootingMode in MAID3-COMMON(E).PDF)

9. ContinuousShootingNum
This will set the number of shots in continuous shooting by host.
(refer to kNkMAIDCapability_ContinuousShootingNum in MAID3-COMMON(E).PDF)

10. RemainContShooting
This will get the number of shot that can be recorded on SDRAM or the card in the continuous shooting mode by the command.
(refer to kNkMAIDCapability_RemainContinuousShooting in MAID3-COMMON(E).PDF)

11. ShootingLimit
This will set the shooting limit number in continuous shooting.
(refer to kNkMAIDCapability_ShootingLimit in MAID3-COMMON(E).PDF)

12. RemainCountInMedia
This will get the number of shot that can be saved in Card in current image quality.
(refer to kNkMAIDCapability_RemainCountInMedia in MAID3-COMMON(E).PDF)

13. MovieWBTuneColorTempEx
This will set the color temperature when the WBMode is “Color Temperature”. (Video Recording Menu).
(refer to kNkMAIDCapability_MovieWBTuneColorTempEx in MAID3-COMMON(E).PDF)

14. WBTuneColorAdjust
This will set the white balance adjustment when the WBMode is “Color Temperature”. (Photo shooting menu).
(refer to kNkMAIDCapability_WBTuneColorAdjust in MAID3-COMMON(E).PDF)

15. MovieWBTuneColorAdjust
This will set the white balance adjustment when the WBMode is “Color Temperature”. (Video Recording Menu).
(refer to kNkMAIDCapability_MovieWBTuneColorAdjust in MAID3-COMMON(E).PDF)

21. SpecialShootingMode
Show the status of special shooting mode.
(refer to kNkMAIDCapability_SpecialShootingMode in MAID3-COMMON(E).PDF)

22. Slot2JpegImageSize
(refer to kNkMAIDCapability_Slot2JpegImageSize in MAID3-COMMON(E).PDF)

4. Live View

You can change Live View settings. Sub menu will be display as in below list.
4)
Select the item you want to set up
1. LiveViewProhibit 2. LiveViewSelector
0. Exit

1. LiveViewProhibit
Get the Live view prohibition condition.
(refer to kNkMAIDCapability_LiveViewProhibit in MAID3-COMMON(E).PDF)

2. LiveViewSelector
This will set live view photography or movie live view.
(refer to kNkMAIDCapability_LiveViewSelector in MAID3-COMMON(E).PDF)

5. Autofocus

You can change Autofocus settings. No sub menu.
5)
No sub menu.

※Mirrorless cameras do not support this feature

6. Capture

Take a picture. If you try to transfer images to PC, you must change 11)SaveMedia of Sub Menu 3 to SDRAM. When you set it to Card, the module does not notify of the addition of the item object.
(refer to kNkMAIDCapability_Capture in MAID3-COMMON(E).PDF)
6)
Select (1-6, 0)
Select the item you want to set up
1. Single Shooting 2. Continuous Shooting
3. Interval Shooting 4. SelfTimer Shooting
5. Bulb Shooting 6. IntervalBulb Shooting
0. Exit

7. CaptureAsync

Take a picture. The result of this Capability is confirmed by DeviceReady. If you try to transfer images to PC, you must change 11) SaveMedia of Sub Menu 3 to SDRAM. When you set it to Card, the module does not notify of the addition of the item object.
7)
Select (1-8, 0)

Select the item you want to set up

1. Single Shooting 2. Continuous Shooting
3. Interval Shooting 4. SelfTimer Shooting
5. Bulb Shooting 6. Cancel Shooting (※)
7. BulbInterval 8. Focus Shift
0. Exit

(※) This menu is only supported on Windows.
For the Mac version, the following menu will be displayed after starting a capture that can be cancelled.
Select (1, 0)
1. Cancel Shooting 0. Exit

8. AFCaptureAsync

This will take a picture after auto focus. The result of this Capability is confirmed by DeviceReady. If you try to transfer images to PC, you must change 11)SaveMedia of Sub Menu 3 to SDRAM. When you set it to Card, the module does not notify of the addition of the item object.

8)
Select (1-7, 0)
Select the item you want to set up

1. Single Shooting 2. Continuous Shooting
3. Interval Shooting 4. SelfTimer Shooting
5. Cancel Shooting (※)
6. BulbInterval 0. Exit

(※) This menu is only supported on Windows.
For the Mac version, the following menu will be displayed after starting a capture that can be cancelled.
Select (1, 0)
1. Cancel Shooting 0. Exit

9. AFCapturesync

This will take a picture after auto focus and save an image to specified media.
Select (1-6, 0)
Select the item you want to set up

1. Single Shooting 2. Continuous Shooting
3. Interval Shooting 4. SelfTimer Shooting

0. Exit

10. SB Menu

You can change flash settings. Sub menu will be display as in below list.
10)
Select the item you want to set up

1. InternalFlashComp 0. Exit

1. InternalFlashComp
Set the flash compensation amount.
(refer to kNkMAIDCapability_InternalFlashComp in MAID3-COMMON(E).PDF)

11. MovieMenu

You can change movie settings. Sub menu will be display as in below list.
11)
Select the item you want to set up

2. CountUpMethod	3. ExternalRecordingControl	
4. Active_D_Lighting	5. AfAreaMode	6. AFSubjectDetection
7. Attenuator	8. CCDDataMode	9. FileType
10. FocusMode	11. PictureControl	12. MovieSkinSoftening
13. VibrationReduction	14. ISOAutoHiLimit	15. MeteringMode
16. H265ToneMode	20. PortraitImpressionBalance	21. RecMicrophone
22. RecMicrophoneValue	23. RecordingZone	24. ScreenSize
25. ShutterMode	26. WindNoiseReduction	0. Exit

2. CountUpMethod
Set “Timecode - Count-up method” on movie shooting menu.
(refer to kNkMAIDCapability_CountUpMethod in MAID3-COMMON(E).PDF)

3. ExternalRecordingControl
Set “HDMI- External recording control”.
(refer to kNkMAIDCapability_ExternalRecordingControl in MAID3-COMMON(E).PDF)

4. Active_D_Lighting
This will set Active D-Lighting. (Photo shooting menu).
(refer to kNkMAIDCapability_MovieActive_D_Lighting in MAID3-COMMON(E).PDF)

5. AfAreaMode
Set AF-area mode for movie.
(refer to kNkMAIDCapability_MovieAfAreaMode in MAID3-COMMON(E).PDF)

6. AFSubjectDetection
Set “AF subject detection options”
(refer to kNkMAIDCapability_MovieAFSubjectDetection in MAID3-COMMON(E).PDF)

7. Attenuator
Set attenuator for movie.
(refer to kNkMAIDCapability_MovieAttenuator in MAID3-COMMON(E).PDF)

8. CCDDataMode
(refer to kNkMAIDCapability_MovieCCDDataMode in MAID3-COMMON(E).PDF)

9. FileType
Set the movie file type.
(refer to kNkMAIDCapability_MovieFileType in MAID3-COMMON(E).PDF)

10. FocusMode
Set focus mode for movie.
(refer to kNkMAIDCapability_MovieFocusMode in MAID3-COMMON(E).PDF)

11. PictureControl
Set and get current Picture Control for movie.
(refer to kNkMAIDCapability_MoviePictureControl in MAID3-COMMON(E).PDF)

12. MovieSkinSoftening
This represents the "beautiful skin effect" option in the video recording menu.
(refer to kNkMAIDCapability_MovieSkinSoftening in MAID3-COMMON(E).PDF)

13. VibrationReduction
Set vibration reduction for movie.
(refer to kNkMAIDCapability_MovieVibrationReduction in MAID3-COMMON(E).PDF)

14. ISOAutoHiLimit
This represents the "Movie ISO Sensitivity Setting - Maximum Control Sensitivity" in the video shooting menu.
(refer to kNkMAIDCapability_MovieISOAutoHiLimit in MAID3-COMMON(E).PDF)

15. MeteringMode
Set the metering mode of camera.
(refer to kNkMAIDCapability_MovieMeteringMode in MAID3-COMMON(E).PDF)

16. H265ToneMode
Set “Video file type” - “H.265 10-bit(MOV)” - “Tone mode”.
(refer to kNkMAIDCapability_MovieH265ToneMode in MAID3-COMMON(E).PDF)

20. PortraitImpressionBalance
This represents the "Portrait Impression Adjustment" option in the video recording menu.
(refer to kNkMAIDCapability_MoviePortraitImpressionBalance in MAID3-COMMON(E).PDF)

21. RecMicrophone
This will set the value of Microphone. (Video Recording Menu).
(refer to kNkMAIDCapability_MovieRecMicrophone in MAID3-COMMON(E).PDF)

22. RecMicrophoneValue
This will set the microphone sensitivity when kNkMAIDCapability_MovieRecMicrophoneValue is manual sensitivity. (Video Recording Menu).
(refer to kNkMAIDCapability_MovieRecMicrophoneValue in MAID3-COMMON(E).PDF)

23. RecordingZone
This will set " Frequency response". (Video Recording Menu).
(refer to kNkMAIDCapability_MovieRecordingZone in MAID3-COMMON(E).PDF)

24. ScreenSize
Set frame size / frame rate for movie.
(refer to kNkMAIDCapability_MovieScreenSize in MAID3-COMMON(E).PDF)

25. ShutterMode
(refer to kNkMAIDCapability_MovieShutterMode in MAID3-COMMON(E).PDF)

26. WindNoiseReduction
This will set "Wind noise reduction". (Video Recording Menu).
(refer to kNkMAIDCapability_MovieWindNoiseReduction in MAID3-COMMON(E).PDF)